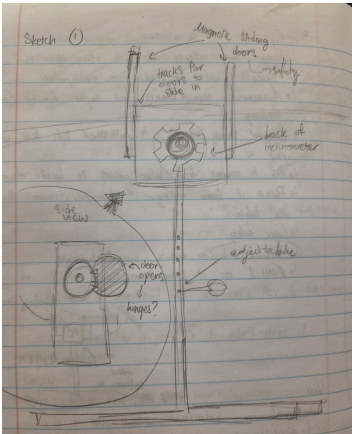
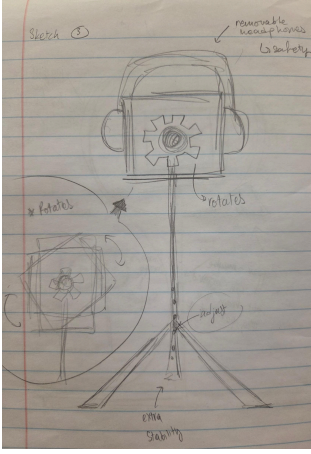
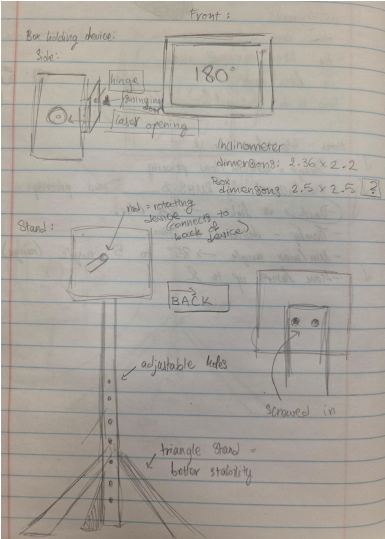
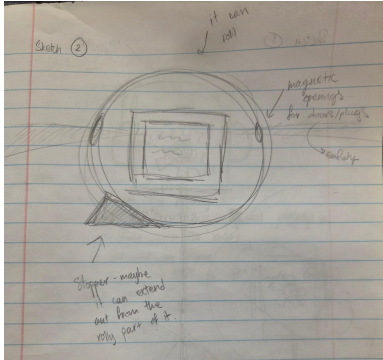
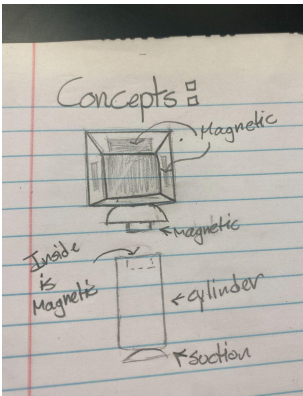
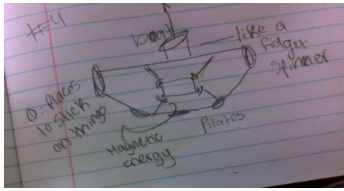
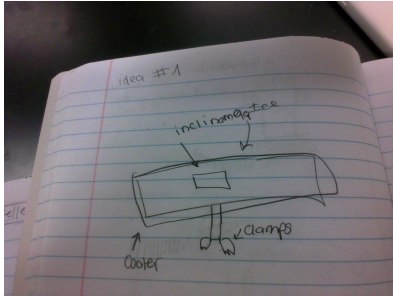
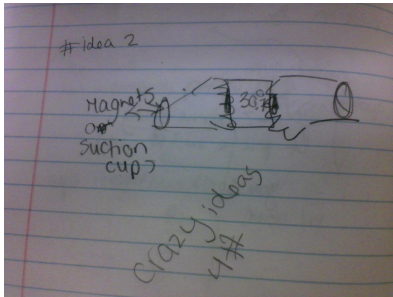
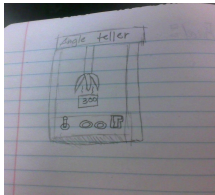
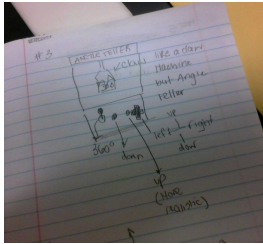
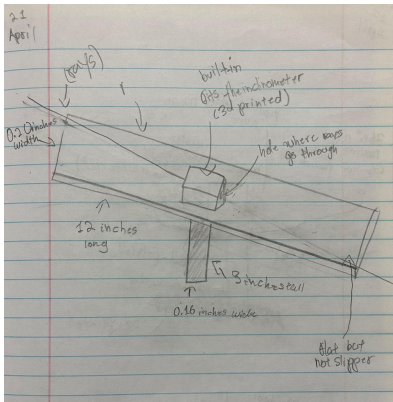
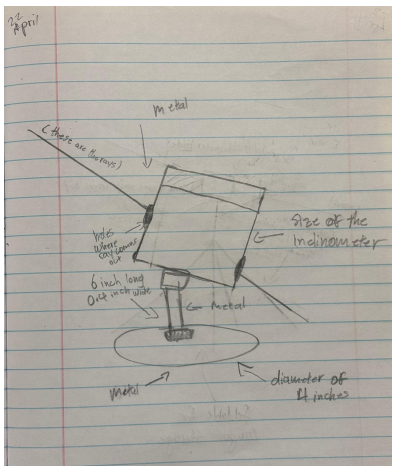


Idea/concept for the project	Unique details about this project	Can/should it be built?	Date
1. 	• Magnetic sliding doors allow for protection from lasers ◦ May be difficult to build considering the budget • The gear system in the back allows for 360-degree rotation ◦ May be difficult to find a device that allows for this • Adjustable stand	Might be difficult to build, no.	4/24/26
2. 	Pros: • Some kind of adjustable/removable headphone system is a safety feature • Gear system in the back • Adjustable stand Cons: • May be difficult to find or create the gear system	Might be difficult to build, no.	4/24/26
3. Chosen design 	Pros: • A metal/wooden rod attached to the back of the main box allows for the entire structure to rotate 360 degrees • Legs are in a triangular shape - the most sturdy shape for better stability • Holes in the main stand allow for direct adjustability/device can move up and down • A hinged door and a potential latch allow for the door to swing and protect from harmful lasers • The device is inside a box • Inside the box is magnetic so that the device is stable Cons: • Lining the inside of the box with the magnets may be difficult	Yes	4/24/26
4.	Pros:	No	4/24/26

	• There is a magnetic opening as a safety feature for the device • A stopper to prevent the device from rolling • The round shape allows for 360-degree rotation Cons: • Not very adjustable height-wise • Kind of inconvenient/could be lost easily • Stopper might look inconsistent		
5. 	Pros: • Able to put together easily. • The sphere can move 360 degrees in the cup - Making it easier to get an angle from all directions. • has multiple suction cups on the bottom of the leg stands that make it more stable. • A thick layer around where the device would be put, making it harder to break it. Cons: • If turned the wrong way, one of the laser holes could be covered. • The cup we put the sphere in could lose magnetism. • Suctions may work on certain surfaces.	Could potentially be built, but it shouldn't be built right now.	4/24/26
6. 	Pros: • Can pull apart and put into storage easily. • Magnets on each wall are designed so that the device cannot fall out easily. • 1 big suction on the bottom that can be stuck on linoleum or a whiteboard. • Doors on the side of the box that cover the lasers, which can open and close. Cons: • If the bottom part is lost, the students can't use the device.	It could be built, but it shouldn't be built because it is not effective.	4/24/26

	It is not adjustable and cannot be used on every surface.		
7. 	Pros - Magnets/suction cups can be taken apart (Two different options depending on the surfaces) or a sticky surface - It's a strong boat-like structure	Not ideal, as suction cups are not stable and magnets are hard to come by.	22nd April, 2026
	Pros - Lasers will be seen through the ice but will not damage the eyes. (I think) - When you successfully get an angle correct, Stateside starts to play - The ice prism protects the inclinometer. Cons - Not very stable, with only one clamp holding the whole ice prism and cooler outline. - Not very useful without the cooler, the ice will melt. - The inclinometer can only face up, or else you won't see the angle. - Very slippery grip.	Yes, is possible because all the materials are easy to come by	22nd April, 2026
	Pros - Magnets/suction cups can be taken apart (Two different options depending on the surfaces) - A hand-like structure to resemble a human's grip. - Easy to make/cheap to make - The hands = laser shield Cons - Can only be placed horizontally (<----->) - Without the hands, laser injuries will be more frequent.	No, as the hands are not easy to build/design online or in person.	22nd April, 2026

 	Pros - Works like a claw machine, except no toys inside :(. - Easy to handle - Acts just like a claw machine; Able to move up and down, left and right, and 360 Degrees. - No glass = Easy access - The claw is used as the Laser shield - The claw has magnets to easily attach itself to the Inclinometer. - Has an eye-catching/attention-catching name to it (THE ANGLE TELLER) Cons - Very big for such a small tool - Could be very expensive. - Would need to buy a claw machine and empty it all, or use electrical robotics to make your own.	Not ideal, as the claw machine would not be easily adjustable/easy to move.	April 24th, 2026
 (Forgot to add: can adjust the t, works like a 'manual seesaw')	Pros: - Simplistic approach that doesn't require a complicated production process - Can be easily used by anyone - Nothing needs to be set up besides Cons - The length of the rectangle that holds the inclinometer might make it harder to see where it's going - Restricted in its movement - May be hard to identify because of how generic it is	Yes, it can be made using basic 3D shapes, whether with materials such as wood or metal, and it only requires basic 3D printing skills.	21st April 2026
	Pros: - The wide metal base and rigid pot resist wobble, which is ideal for accurate angle readings - Not a lot of moving parts make it easier to produce - Long-lasting materials Cons - Heavier due to most of it being metal - Works well for one measurement setup, but not ideal for varied environments - You can't easily adjust the height or angle	No, it can't be made because of how many metal parts are critical for the production; this would need tools that aren't available to us.	22nd April 2026

	without adding mechanisms		
	Pros - Foldable legs and lightweight material make it easier to carry around and set up - Height and angle can be quickly changed, making it useful for many angle measurement scenarios - Tripod legs can adapt to terrain better than a flat base Cons - A 3D-printed holder made of plastic will likely result in micromovements (sliding around), which affects the precision - Tripods can shift slightly if the legs aren't locked tightly - More likely to fall apart due to light weight nature of the overall design	It is buildable, however, figuring out how to make the stand and holder connect may require a difficult mechanism to engineer	22nd April

D2 Chosen Design:

D3 - Plan of Action: (build plans)

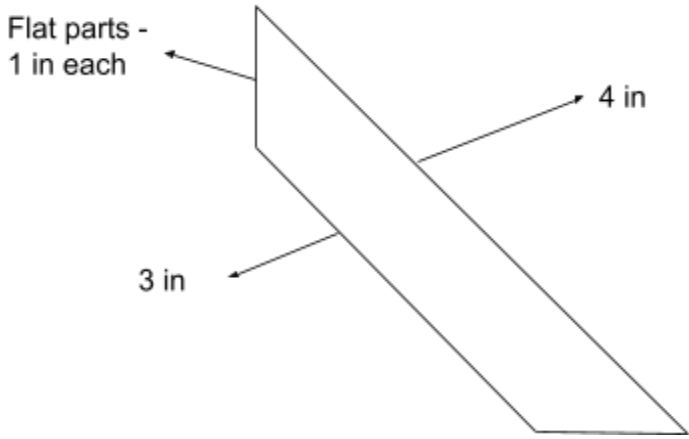
Things to do before:

- Onshape the box in which the inclinometer will be sitting
- Onshape ideas: The box attachment will have a plastic rod attached to a removable gear system that we can manually turn to move the rotating system.

Onshape:

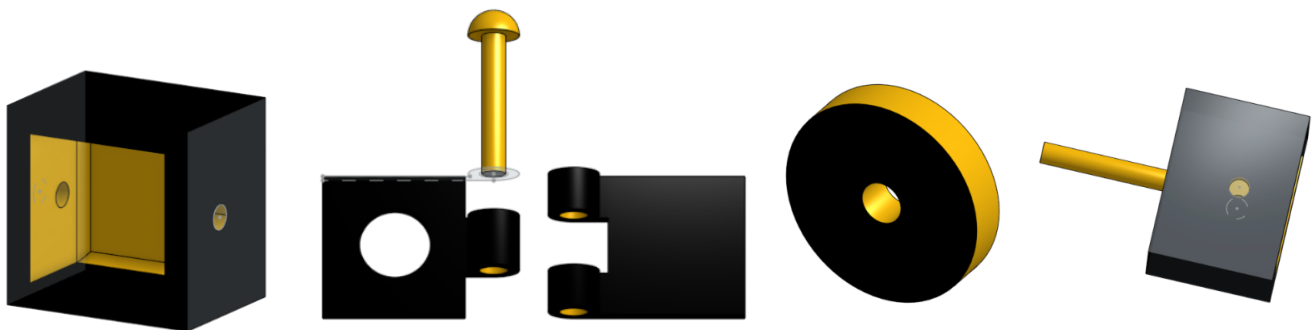
- [Here](#) is our Onshape box and gear mechanism design.
- [YouTube tutorial](#) for Hinge!

5/1 - Day 1	[Print out Onshape design] Begin designing the stand - Use a 2x4 piece of wood - Suggested length: 11 in tall and 2 in ish thick
5/4 - Day 2	[Print out Onshape design and/or make edits] Continue making the stand and begin working on the legs if time - Leg information: (out of wood) - Flat top and bottom - One side is 3-ish in - The other side is 4-ish in (picture below)

	
5/5 - Day 3	Work on the stand and legs
5/6 - Day 4	Work on the stand and legs
5/7 - Day 5	Work on the stand and legs
5/8 - Day 6	Finish putting the laser covers on the box.
5/11 - Day 7	Assemble the pieces.
5/12 - Day 8	Work on fixing and making sure the final product works
5/13 - Day 9	Touch up the final product
5/14 - Day 10	Finish

D4 - Design blueprint:

First designs for our prototype on Onshape:



Here are some screenshots of our Onshape designs; the full designs can be found by following this [link](#).

The drawing we chose for our final prototype design:

